

* Eigenvalue / eigenvector

- $n \times n$ matrix A , non-zero vector v

$$Av = \lambda v$$

$\lambda \rightarrow$ eigenvalue

* Affine transform

- 선형 변환과 이동의 표현 $\times$, 이동의 표현

$$\begin{bmatrix} x_{\text{new}} \\ y_{\text{new}} \end{bmatrix} = A \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$\downarrow$ type transform $\rightarrow$ bias

$$\begin{bmatrix} x_{\text{new}} \\ y_{\text{new}} \\ 1 \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} & b_1 \\ A_{21} & A_{22} & b_2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \rightarrow \text{Affine transform}$$

1. General Affine transform

$$T = \begin{bmatrix} A & t \\ 0^T & 1 \end{bmatrix} \Rightarrow T^{-1} = \begin{bmatrix} A^{-1} & -A^{-1}t \\ 0^T & 1 \end{bmatrix}$$

2. 강체 변환 (회전 회전 $RESO(3)$ + 이동 (translation))

$$T = \begin{bmatrix} R & t \\ 0^T & 1 \end{bmatrix} \Rightarrow T^{-1} = \begin{bmatrix} R^T & -R^T t \\ 0^T & 1 \end{bmatrix}$$

$SO(n)$: Special Orthogonal Group

$\rightarrow$ n 차원 rotation 만을 다루는 group ($n \times n$)

$$SO(n) = \{ R \in \mathbb{R}^{n \times n} \mid R^T R = I, \det(R) = 1 \}$$

$$\text{Dim(DOE)} : \frac{n(n-1)}{2}$$

$SE(n)$: Special Euclidean Group

→ n 개의 rotation + translation, other pose matrix group $(n+1 \times n+1)$

$$SE(n) = \left\{ \begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix} \mid R \in SO(n), t \in \mathbb{R}^n, 0 \in \mathbb{R}^n \right\}$$

$$\text{DOF} : \frac{n(n-1)}{2} + n = \frac{n(n+1)}{2} \quad (\frac{n(n-1)}{2} \text{개의 회전} + n \text{개의 이동})$$

$$SE(n) = \mathbb{R}^n \rtimes SO(n)$$

$$\det(A) = 6 \times 2 + (-2) \times -3 + 2 \times 4$$

$$= 12 + 6 + 8$$

$$= 26$$

$$\det(A_2) = 5 \times (-3) + 14 \times 4 + 6 \times 11$$

$$= -15 + 56 + 66$$

$$= 107$$

$$\frac{138}{15}$$

$$\det(A_3) = -5 \times 2 + -17 \times 1 + 8 \times 11$$

$$= -10 - 17 + 88 = 61$$

$$\det(A_4) = -14 \times 2 - 17 \times (-3) + 26 \times 11$$

$$= -28 + 51 + 286$$

$$= 309$$

